Román Alejandro López Ruvalcaba

Data Engineer | Analytics Engineer

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PROFESSIONAL SUMMARY

Data engineer with 2+ years in analytics and data engineering. Built and operate in production a Medallion (Bronze/Silver/Gold) data warehouse on SQL Server that integrates SAP ECC, Oracle and an Odoo CRM (an e-commerce source is being added), feeds Power BI reports used daily by the Credit & Collections team, and includes data-quality checks and full documentation. Strong in SQL/T-SQL, dimensional modeling and ETL/ELT; hands-on with dbt and Docker in a public portfolio project (Airflow orchestration in progress), and extending into PySpark, Databricks and Azure.

TECHNICAL SKILLS

Data Engineering: Medallion architecture, ETL/ELT, incremental loads (MERGE), historical backfills, SCD Type 1/2, star schema, data quality & reconciliation, job scheduling (SQL Server Agent)

Languages: SQL / T-SQL (stored procedures, window functions, indexing), Python (pandas, pyodbc)

Sources: SAP ECC (FI/SD), Oracle, PostgreSQL (Odoo), linked servers

BI: Power BI (DAX, Power Query, PBIP/TMDL), semantic modeling, Excel (advanced)

Tools: dbt Core (models, tests, snapshots, macros, docs), Docker, Apache Airflow (hands-on, in progress), Git / GitHub, SSMS, draw.io

Learning: PySpark, Databricks, Azure (DataCamp Associate Data Engineer in SQL, 87%)

WORK EXPERIENCE

Analytics Engineer | CIOSA Autotodo *Aug 2025 – Present*

- Designed and operate a production Medallion data warehouse on SQL Server integrating 3 sources (SAP ECC, Oracle, Odoo/PostgreSQL) through linked servers: 32 raw tables loaded daily by 4 SQL Agent jobs; the Gold layer runs only if all sources loaded successfully.
- Built the load framework: stored procedures, incremental MERGE loads, separate historical backfills, and step-level logging (rows, duration, failing line) with rollback on error.
- Modeled a star schema with SCD Type 1 and Type 2 customer dimensions, rebuilding customer history from the SAP change log instead of projecting current attributes into the past.
- Developed a payment-to-invoice application model that links 96.5% of payment dollars to their invoices and records the reason for the remainder, enabling collections and DSO analysis.
- Removed repeated manual data preparation by exposing curated Gold-layer tables and star-schema views, and improved report load times by moving transformations out of Power BI into the warehouse.
- Created a data-quality layer that generates work lists for fixing master data in SAP; validated a rebuilt receivables-aging model against the original source view.
- Delivered 5 Power BI reports connected to the warehouse (models versioned in Git as PBIP/TMDL). They are in production, answering business questions that several users check every day. Documented every layer.

Data Analyst | NanoTess *May 2024 – May 2025*

- Built Power BI dashboards and Excel models to monitor clinical and operational KPIs for international stakeholders.
- Cleaned, transformed and analyzed trends in treatment data using SQL and Python.
- Automated recurring reporting workflows, reducing manual data entry and preparation for internal teams.

PROJECTS (GitHub: romloc11)

Order-to-Cash Data Warehouse (dbt, Docker, Airflow): anonymized re-creation (synthetic data) of a production warehouse: 3 sources on 3 engines (SQL Server, PostgreSQL, MariaDB), Bronze/Silver/Gold with 26 dbt models and 98 tests, SCD Type 2 rebuilt from a change log, payment-to-invoice bridge, daily receivables snapshot, dbt snapshots, macros and generated docs; Docker Compose. Airflow orchestration in progress. github.com/romloc11/order-to-cash-dwh

EDUCATION & CERTIFICATIONS

- B.Eng. Nanotechnology Engineering, ITESO (2020 – 2025)
- Google Data Analytics Professional Certificate; CS50's Introduction to Programming with Python
- In progress: DataCamp Associate Data Engineer in SQL; Microsoft PL-300

Languages: Spanish (native), English (B2)